Project Intelligence: system architecture

Team chat that turns project conversations into tasks, progress, health and risks. Self-hosted: no external AI, no WhatsApp or Meta API.

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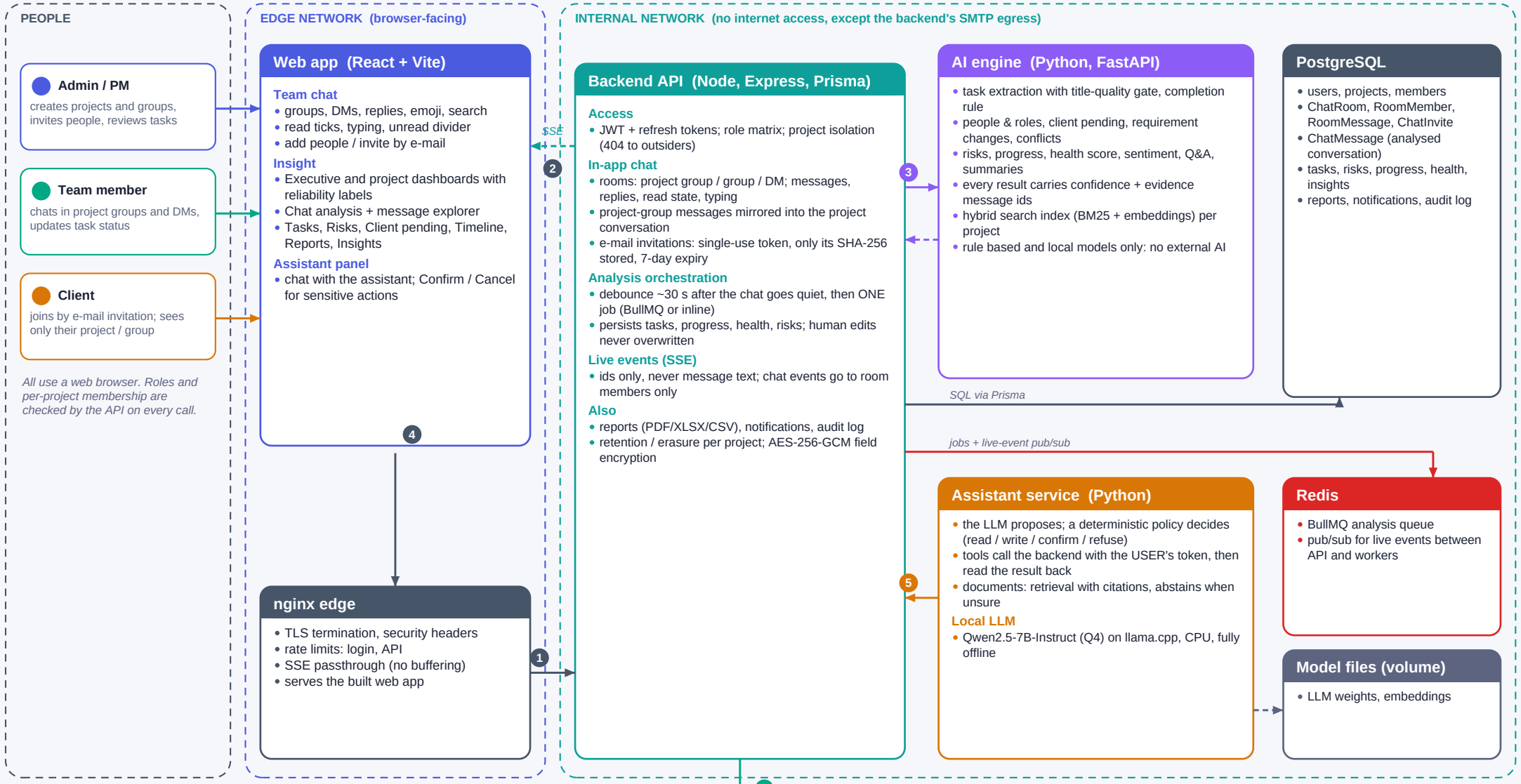
Facts: backend/src, ai-engine/app, assistant/, docker-compose.yml



Guarantees

- Direct messages are private: never analysed, not readable even by admins.
- Every AI result links back to the exact source messages (evidence).
- Human corrections survive re-analysis; thin evidence is labelled LOW confidence.
- Database, queue and AI engine have no internet access; no external AI or WhatsApp.
- Secrets only in the environment; tokens hashed; sensitive fields encrypted at rest.
- Retention and erasure per project also remove derived data and chat copies.

SMTP server (e.g. Gmail app password)

Only outbound connection: invitation and "added to group" e-mails (egress network)

Main flows

1. A message is sent in Team chat; the API stores it (and mirrors it for project groups).
2. Members get a live event (ids only) and refetch; ticks turn blue when everyone has read.
3. After ~30 s of quiet one analysis job runs; the AI engine returns tasks, risks, progress, health.
4. Dashboards show the numbers with evidence and a reliability label (HIGH / MEDIUM / LOW).
5. The assistant answers and acts through the API with the user's own permissions.
6. Invitations e-mail a single-use link; accepting creates a client account in that group.

Production: docker compose (postgres, redis, ai-engine, backend, worker, assistant, nginx edge). Development: scripts/dev/dev-up.ps1.